

ORIGINAL RESEARCH

Violence 4D: Violence detection in surveillance using 4D convolutional neural networks

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Abstract

As violence has increased around the world, surveillance cameras are everywhere, and they are only going to get more ubiquitous. Due to the massive volume of video footage, automatic activity detection systems must be used to create an online warning in the event of aberrant activity. A deep learning architecture is presented in this study using four-dimensional video-level convolution neural networks. The proposed architecture includes residual blocks that are used with three-Dimensional Convolution Neural Networks 3D (CNNs) to learn long-term and short-term spatiotemporal representation from the video as well as record inter-clip interaction. ResNet50 is used as the backbone for three-dimensional convolution networks and dense optical flow for the region of interest. The proposed architecture is applied on four benchmarks for violence and non-violence videos, which are commonly used for violent detection. It obtained test accuracies of 94.67% on RWF2000, 97.29% on Crowd violence, 100% on Movie fight and 100% on the Hockey Fight dataset. These results outperform the previous methods used on RWF2000 datasets.

KEYWORDS

computer vision, video surveillance

1 | INTRODUCTION

Surveillance Cameras are increasing in many places at several countries. They have done many good things. As it is a way for organisations, police, and many other places to keep monitoring people who have bad intentions.

Surveillance cameras help to ensure safety by detecting violence, which is on the rise nowadays and cannot be detected manually. In computer vision society, violence detection is growing steadily, gaining a lot of interest as surveillance cameras can collect evidence and identify possible crimes of violence. However, personal monitoring of a significant volume of video footage in real-time is extremely expensive; because of this computer vision society found that automatically distinguishing crime scenes from recordings has become necessary. The term 'video violence detection' is a subclass of human action recognition that focuses on detecting violent actions from data of video identifying typical human actions. Video data has more temporal sequences than static photos because of the strong

inter-frame correlation, a series of frames that appear one after the other might depict a constant movement.

Because surrounding frames include a lot of redundant information, many scientists focus their attention on appropriately fusing spatial and temporal data. There are primarily two types of approaches for recognising human motion in videos and detecting violence; the first type is traditional feature extraction methods; shallow classifiers are used for feature extraction [1]. Researchers are attempting to create a powerful set of video descriptors that can be fed into a linear classifier.

However, some traditional methods have the disadvantage of being linked to the locations of major points across which data is ignored. Lack of workplace flexibility is also an issue, and computational cost is quite high. The second type for detecting human activity in films utilised by researchers [2] is end-to-end deep learning methods. In latest years, many end-to-end deep learning-based techniques for activity recognition have been presented as they produce better accuracy in general. For some deep learning techniques, there is a problem with feature

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extraction in complex scenes, making it difficult to capture some of the most important features from the entire video. Most 3-Dimensional Convolutional neural networks only explore a small portion of the entire video. Also, during training, clip-based models frequently disregard the video-level structure and long-range spatiotemporal dependence. As a result, a violence detection using 4D Convolutional Neural Networks to detect violence in videos is presented, which called Violence 4D. Because of using four-dimensional blocks on the entire video helps to capture inter clip interactions. This improves representation of three-dimensional convolutional neural networks.

The main contribution of this study is the approach for detecting violence in videos. To model long-term dependence more efficiently, we used video-level representation learning. Violence 4D is divided into two parts: (1) a holistic sampling technique that uniformly samples a succession of short-term units over the whole video, and (2) a 4D convolutional interaction that models long-range spatial-temporal dependence by employing 4D residual block to capture interaction of inter-clip. Violence 4D achieved better performance than previous studies on four violence benchmarks. Namely RWF2000 [45], Movie fight [3], Hockey Fight [4], and Crowd violence [5].

The rest of paper is organised as follows, Section 2: related works for violence detection, Section 3: proposed model violence 4D, Section 4: datasets descriptions and experiments, Section 5: discussion and Section 6 is the conclusion and future work.

2 | RELATED WORKS

In this section, existing state-of-the-art techniques for detecting violence involving traditional methods, and deep end-to-end learning are summarised. Deep learning approaches, which have generated superior outcomes in general, are given special consideration, making deep learning the most used method. The following is a list of traditional methods like spatiotemporal descriptors and deep learning approaches for recognising action and violence in video.

2.1 | Traditional approaches

2.1.1 | Spatiotemporal descriptors

Traditional techniques depend on Spatiotemporal descriptors and visual features produced from methods based on the self-organising map of Kohonen to detect violence from blood [6] and motion descriptors [7]. Initialisation, tracking, posture estimation, and recognition are presented as different hierarchies for recognising actions [8]. Moreover, ref. [9] focus solely on whole-body motions, excluding studies on gesture recognition. In ref. [10], the authors considered Space-Time Interest Points, Scale-Invariant Feature Transform (SIFT), and Motion Scale-Invariant Feature Transform methods as spatiotemporal descriptors, utilising a bag-of-features to describe each video and Support Vector Machine (SVM) to categorise it with

separate kernels. Other space-time descriptors, such as Fisher vectors [11], instead of utilising points, characterise each video using a set of characteristics based on vectors of concern.

The computational cost of extracting spatiotemporal characteristics in bag-of-features frameworks is quite high, making them unsuitable for real-time applications. Different space-time descriptors have the disadvantage of being linked to the locations of major points across which data is ignored. Lack of workplace flexibility is also an issue when dealing with situations like camera movements, variations in appearance in the same class, and dynamic background; this reduces their output on actual videos.

2.1.2 | Optical flow-based features

After using spatiotemporal descriptors, new features such as Lagrangian direction fields [12] have been investigated. Low contrast characteristics, such as the optical flow local histogram [13], the directed gradient local histogram [14], and the absolute difference between successive frames and features from motion blobs [15]. Other approaches [5] concentrate solely on the optical flow. The Gaussian model of optical flow is utilised to generate areas that are used to create a descriptor which is categorised using SVM. The histogram of optical flow intensity and direction, is a new feature descriptor, were added to this method [16].

Optical flow-based techniques for detecting coherent motion descriptors of moving objects have gained a lot of traction. Ref. [17] used the optical flow to produce densely tracked sampled locations, which were then used to describe human actions. Even though it is an effective way for recognising actions, it necessitates the use of specialised hardware, making it computationally costly. Variance Inflation Factor (VIF) [5] is suggested for real-time detection of violence. However, the direction characteristic of optical flow was not used by VIF. Ref. [18] overcomes this problem by presenting violent Oriented Variance Inflation Factor, a new feature extraction technique, which makes use of statistical movement directions perks of changing statistics on the amount of movement data. After that, the optical flow histogram technique was extended to identify activities in the distance [19]. The technique employs either a histogram of optical flow magnitude or an optical flow direction histogram. This appears to work for a few uncommon situations, like jogging in public or heading the other way, among others. The disadvantage of this is that it is necessary to specify which type of histogram should be utilised for certain functionality. Non-deep learning approaches, which are considered state-of-the-art, use SVM for classification, in feature coding [20], the Fisher vector is used, and Dense Trajectories for spatiotemporal feature extraction [21].

2.2 | Deep learning approaches

In deep learning approaches for the speciality in violence identification, convolutional networks [22] evaluate visual

spatiotemporal characteristics. Long Short-Term Memory (LSTM) layers encode temporal information [23]. To extract spatial information, [24] showed a model that included convolutional, batch normalisation, and pooling layers, as well as minimisation operations. The level variations of the frames or temporal characteristics that define violent situations are then encoded using a recurrent Convolutional Layer Long Short-Term Memory (ConvLSTM). The categorisation is carried out with completely linked layers; at the end, all frames are applied in a certain order. The AlexNet network was utilised to extract features from frames [25], as the CNN model, which has already been pre-trained with ImageNet. Then, a three-component model was proposed [26], consisting of a spatial encoder, a classifier, and a temporal encoder. VGG13 [27], data is fed to the spatial encoder, which is made up of an adjusted CNN, from the subtraction process between frames. As a result, feature maps of spatial characteristics are obtained for each frame, and the temporal encoder. Bidirectional Convolutional LSTM layer receives this data. Using an element-wise maximum approach, the findings are integrated in a representation of the video. In the end, a classifier uses this representation to assess whether the video contains violent activity. Xception Bidirectional LSTM Attention model [28]. This model begins by consistently retrieving 5 and 10 frames from each video. After that, Fight-CNN, a CNN network based on the Xception architecture with a larger kernel size, is used to extract and record spatial characteristics of violent scenarios, because a bidirectional LSTM can figure out how prior and present data relate to one another while considering temporal characteristics. In the end, an attention layer [29] is included to help differentiate essential portions. The attention layer, in combination with bidirectional LSTM layers, decides whether the video contains violence. So far, we've looked at sequential one-stream models. In other words, the input images either have an optical flow or RGB format. On the other hand, other models use a variety of inputs, including both formats and more, which are multi-stream convolutional models, each of which looks at a different type of visual feature.

Fight Net, a multi-stream model with three input modalities [30]. RGB frames, optical flows, and frames of acceleration, each with its own network. Video input is divided into parts, and for each stream, feature maps are obtained for each segment. In the end, all these feature maps have been combined. The network's ultimate output is the average score of each video segment. Fan et al. [31] employed a slow pathway with a low frame rate to capture spatial semantics and a fast pathway with a high frame rate to record motion with precise temporal resolution. Ahmed et al. [32] extracted characteristics from video frames using VGG16, VGG19, and ResNet50, then employed LSTM with an attention layer. Vosta and Yow [33] presented another technique that uses (ResNet50) to extract critical information from each frame of the input stream, followed by a specific schema of recurrent neural networks Convolution LSTM (ConvLSTM) for detecting violence. Certain irregularities occurred while using this technique in the footage, and it could not see anyone in some videos. Ref. [34] showed two-stream CNN; spatial stream

convolution network which is used to learn from RGB images as it acts on individual video frames, executing action detection from still images, while the temporal stream convolution network is utilised to simulate optical flow. The outcomes of the two streams are then combined at a later stage for final prediction. Transfer learning [35] was utilised to recognise violent human behaviour in violent situations. Flow Gated Network presented in ref. [45] makes use of the advantages of 3D convolutional neural networks as well as optical flow. There is a problem with feature extraction in complex scenes, making it difficult to capture some of the most important features from the entire video.

For deep learning, there is a problem with feature extraction in complex scenes, making it difficult to capture some of the most important features from the entire video. 3D CNNs give a huge number of model parameters, which necessitates a huge amount of training data to achieve good results. Furthermore, most 3D CNNs only explore a small portion of the entire video. Also, during training, clip-based models frequently disregard the video-level structure and long-range spatiotemporal dependence.

Several long-term modelling methods for capturing more complicated temporal patterns for learning video-level feature representation have been explored. Wang et al. [36] proposed a Temporal Segment Network which models video-level temporal features using sparse sampling and an aggregation technique. Because of the average aggregate, it is hard to model precise primary information with this technique. To simulate the relationship between temporal segments, Zhou et al. [37] introduced the Temporal Relational Reasoning Network (TRN) by encoding each segment's own representation with related networks TRN can simulate order of video-level temporal information, but it cannot capture deeper temporal information.

3 | PROPOSED METHOD

This section introduces the Violence 4D model for automatic violence detection from video. The Violence 4D major goal is to build a full-fledged network that can identify violence in videos. Violence 4D consists of three main components which are Dense optical flow, ResNet50 and 4D residual blocks as shown in Figure 1. Utilising video four-dimensional convolution neural networks V4D CNN [38]. It processes input of RGB frames from videos with 3D data. As three-dimensional convolution neural networks 3D CNNs takes short term snippet without taking into account the development of 3D spatiotemporal characteristics for level representation in the video. V4D globally models 3D spatiotemporal features using four-dimensional residual blocks as shown in Figure 2. This permits 3D CNNs to be converted to 4D CNNs in order to learn long-range 3D feature interactions and capture inter-clip interactions to increase the representation power of the original clip-level 3D CNNs. By including residual blocks into the current 3D CNNs to perform hierarchically long-range modelling, resulting in higher video level representation into

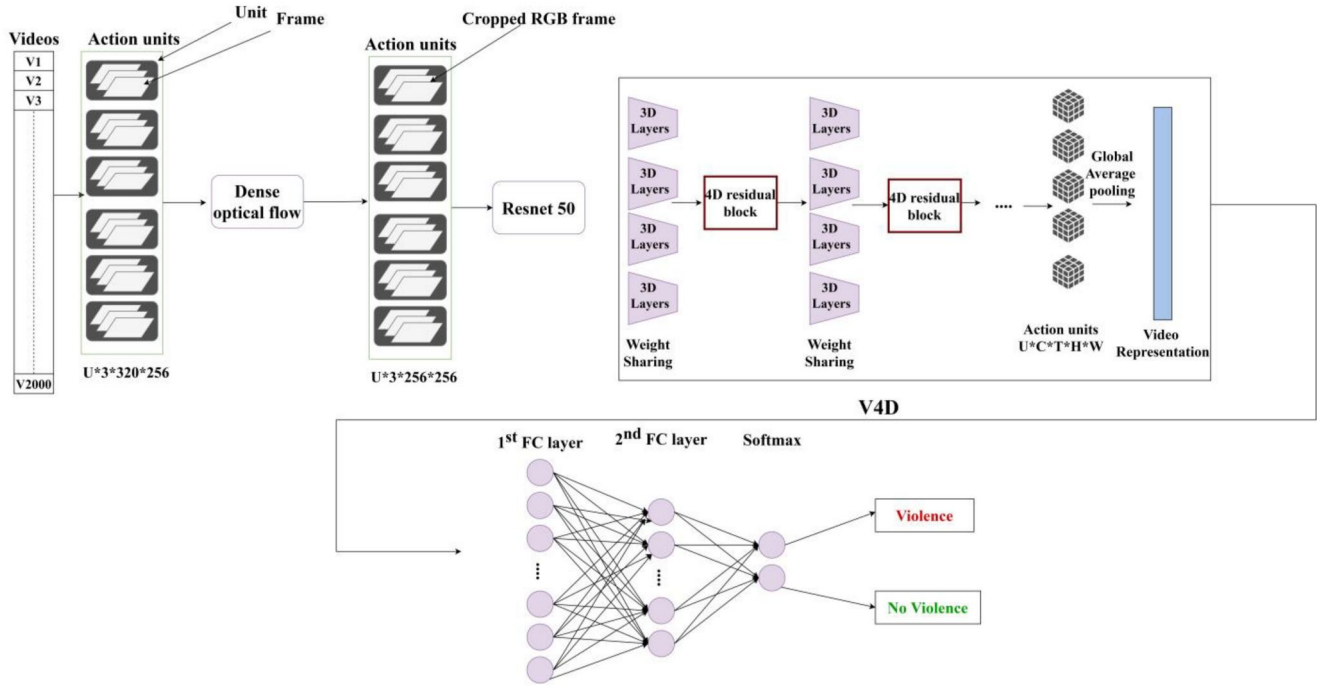


FIGURE 1 Violence 4D architecture for violence recognition which divided into whole sapling for videos and interaction of 4D convolutions on backbone model ResNet50.

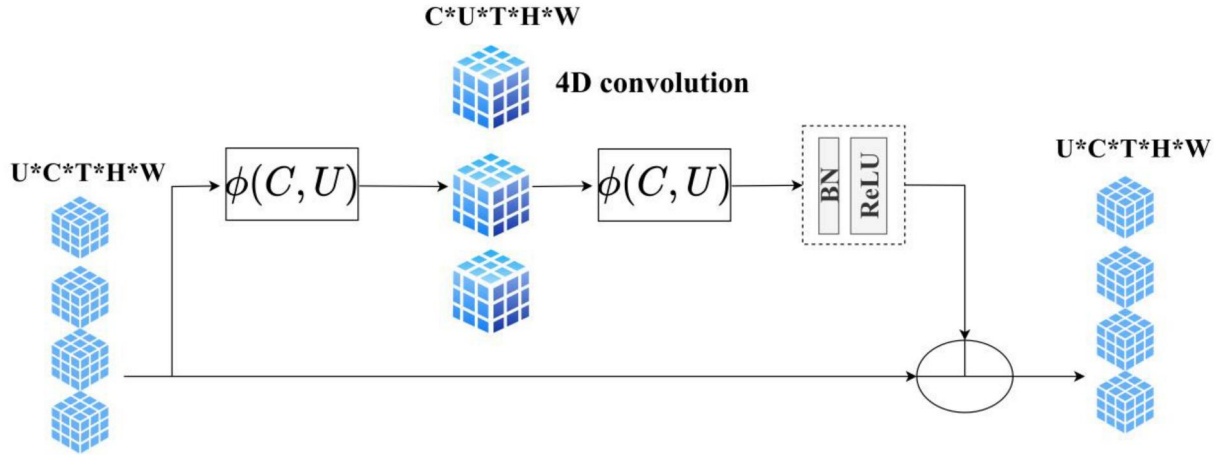


FIGURE 2 4D residual blocks [34].

the current 3D CNNs to do long-range modelling hierarchically. The violence 4D model is divided into three phases namely, Video Sampling, Spatiotemporal Interactions Learning and Training using V4d CNNs Architecture.

3.1 | Video sampling

A video level sparse sampling strategy [39] has been used to get representation for violence recognition, and the input has been included both the overall length of each input video and the short-term action information as it should be preserved. The video is split into U parts throughout (action units); each action

unit contains the sequence of frames; next, choose a sample from each area that demonstrates a short-term action pattern. Then, to depict the entire action in a video, we have sections of action units. Then frames in each section of the action unit are cropped according to the region of interest resulting from using Gunner Farneback's [40] technique for calculating intense optical flow between adjacent frames, as shown in Figure 3. The motion is determined between consecutive frames of a sequence. The norm of each vector has been used to produce a heat map for signalling movement intensity. Then add all the heat maps together to get a final motion intensity map. The region of interest in the image is isolated from the rest, which is the position of the greatest amount of

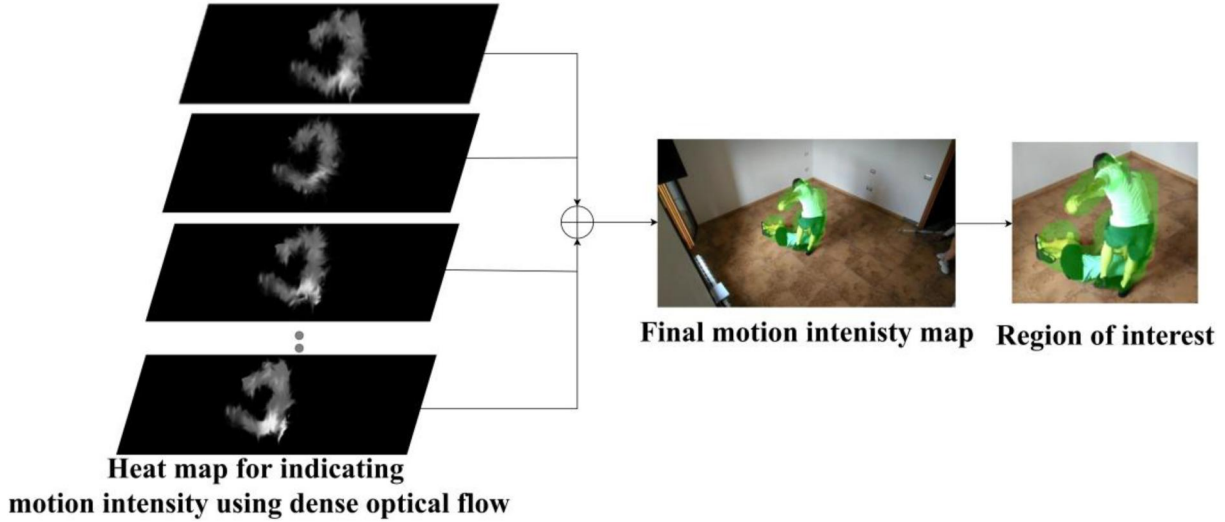


FIGURE 3 Dense optical flow.

movement. After that, each action unit is picked at random from each of the U parts during training. Each action unit's centre is located exactly in the middle of the section that corresponds to it in testing. Consequently, we have the video input $v = \{a1, a2, \dots, au\}$, where $ai \in R^{U \times C \times T \times H \times W}$, while U denotes the number of action units, C denotes the number of available channels, T is the temporal duration, and the action unit's height is H , while its width is W .

3.2 | Learn spatiotemporal interaction

V4D is being used to learn spatiotemporal interactions of 3D convolution kernels that are particularly excellent at modelling short-term spatiotemporal features. Utilising operations that can model both short-term and long-term spatiotemporal representations at the same time. As a result, four-dimensional convolutions have been employed to better describe long-range spatiotemporal interactions. Four-dimensional convolutions are fed a tensor of $(C \times U \times T \times H \times W)$. C denotes the number of available channels; U denotes the number of action units; T denotes the temporal duration; H denotes the height; and W is the width of each action unit.

According to Ref. [31], a pixel in the output at the k th channel's position (u, t, h, w) is denoted as

$$o_j^{utbw} = b_j + \sum_c \sum_{s=0}^{S-1} \sum_{p=0}^{P-1} \sum_{q=0}^{Q-1} \sum_{r=0}^{R-1} W_{jc}^{spqr} U_c^{(u+s)(t+p)(h+q)(w+r)} \quad (1)$$

C represents one of the feature maps' c_{in} input channels. Input and b_j is a bias term. V, S, P, Q , and R is the shape of a 4D convolutional kernel, W_{jc}^{spqr} is the weight at location (s, p, q, r) of the kernel corresponding to the c th channel of the input feature maps, and the j th channel of the output feature maps, and weight at the position (s, p, q, r) of the kernel,

corresponding to the channels. The convolutional operation is linear, and the sequential sum techniques in Equation (1) are interchangeable. Consequently, Equation (2) has been utilised, in which 3D convolutions may be used to create 4D convolutions, allowing the use of 3D convolutions to implement 4D convolutions.

$$o_j^{utbw} = b_j + \sum_{s=0}^{S-1} \left(\sum_c \sum_{p=0}^{P-1} \sum_{q=0}^{Q-1} \sum_{r=0}^{R-1} W_{jc}^{spqr} U_c^{(u+s)(t+p)(h+q)(w+r)} \right) \quad (2)$$

Short-term 3D characteristics of an individual action unit may be represented simultaneously with the long-term temporal development of numerous action units in 4D space using a 4D convolutional kernel. V4D can model movies in a 4D feature space, allowing it to learn more complex relationships between long-range 3D spatiotemporal representations.

3.3 | Training using V4D CNNs architecture

For action recognition, four dimensional convolutions may be incorporated into current 3D CNN architecture. ResNet50 has been used, detailed structure of 3D backbone is shown in Figure 4.

For video-level action detection, the 4D residual block allows collecting both short-term 3D characteristics and long-term development of spatiotemporal representations.

The Residual 4D Convolution Block has the following definition:

$$y_{3D} = x_{3D} + \varphi_{(U,C)} \left(F_{4D} \left(\varphi_{(C,U)} (x_{3D}); W_{4D} \right) \right) \quad (3)$$

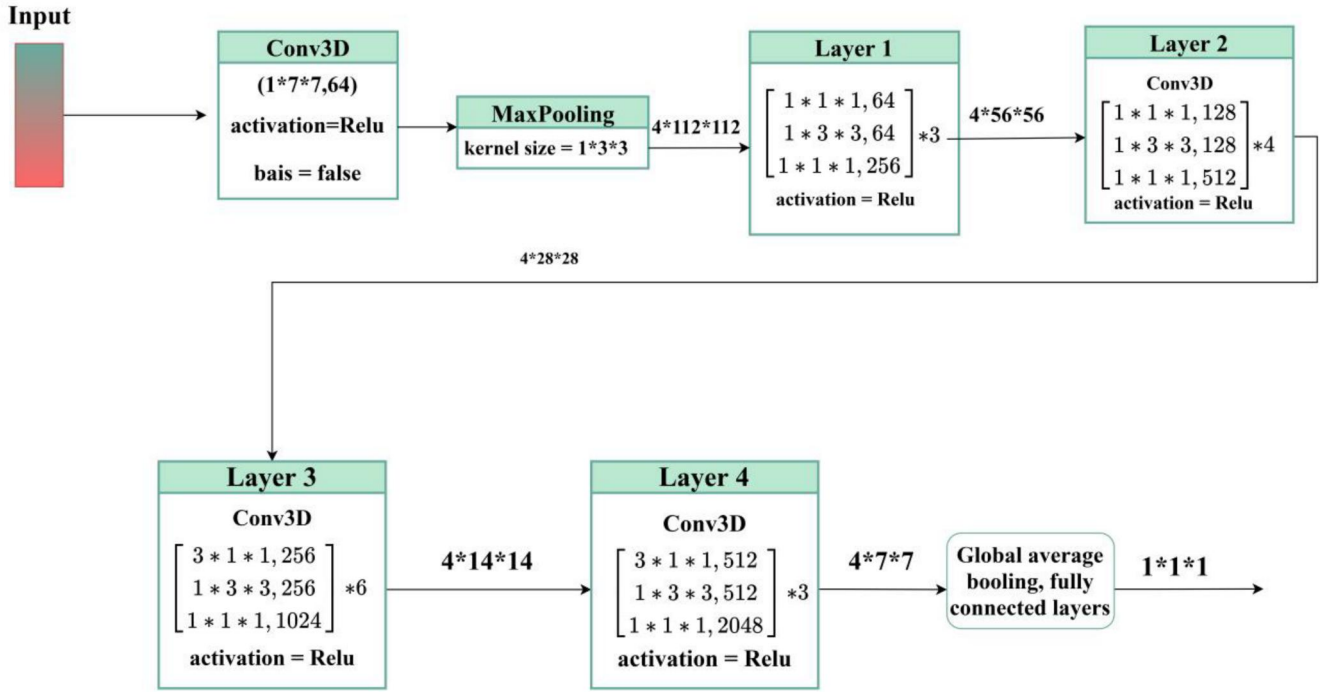


FIGURE 4 13D-slowpath of ResNet50.

The 4D convolution is introduced in Equation (3) as $F_{4D}(X_{3D}; W_{4D})$. $X_{3D}, Y_{3D} \in R^{U \times C \times T \times H \times W}$, a batch dimension is created by combining U so that X_{3D} and Y_{3D} may be immediately processed by standard 3D CNNs. Dimensions can be permuted using a permutation function ϕ . ϕ is used to permute the dimensions of $U \times C \times T \times H \times W$ of X_{3D} to $C \times U \times T \times H \times W$ so that 4D convolutions can handle it. The output dimensions of the 4D convolution are then permuted back to 3D form to match X_{3D} . After that, batch normalisation and ReLU activation [41] are used, as shown in Figures 1 and 2.

The networks' convolutional layers are made up of 3D convolution layers with Residual 4D Blocks in the fourth training. Each action unit is trained independently and parallel in the 3D convolution layers, which have similar parameters. The 3D feature action units are then sent into the Residual 4D Block, which may simulate the succeeding action units' long-term temporal evolution. Finally, global average pooling is a collection of all action units from video representation that is supplied to fully connected layers and softmax for violence detection, as seen in Figure 1.

4 | EXPERIMENTAL METHODOLOGY

The efficiency of Violence 4D in identifying violence from videos on various datasets is discussed in this section.

4.1 | Datasets

The four benchmark datasets used in Violence 4D were those that occur most frequently in research on the identification of

cruel behaviour. They have gained a lot of traction, and they are used to compare different methods for identifying violent acts. The following are the datasets:

RWF2000 [45], although certain video datasets for violence identification exist, they suffer from limitations such as small scale, limited variety, as well as poor imaging resolution. Furthermore, some of the high-quality violence datasets are taken from movies that aren't near enough to real-world scenarios. To address the issue of lack of high-quality data from actual acts of violence, RWF2000 contains 2000 videos collected from YouTube from surveillance cameras in the real world sliced into snippets within 5 s at 30 frames per second. Videos are divided into Violent or Non-violent. None of the videos are modified by multimedia technologies.

Movie dataset [3], here are 200 clips in this collection of sequences from action movies, including fights and non-fighting situations.

Crowd violence [5] is a collection of videos depicting violence. It differs from the others in that it concentrates on the masses rather than on individual conflict. However, it is worth noting that the model's flexibility is documented in 246 videos.

The Hockey Fight dataset [4] is a compilation of fights involving players from the National Hockey League in the United States of America. It contains 1000 videos.

4.2 | Experiments and results

For the Violence Detection Benchmark Datasets, as shown in Table 1, details of datasets are represented. Inside and outside situations, and a variety of meteorological conditions, are

TABLE 1 Details of RWF2000, Hockey, Movie, and Crowd violence datasets

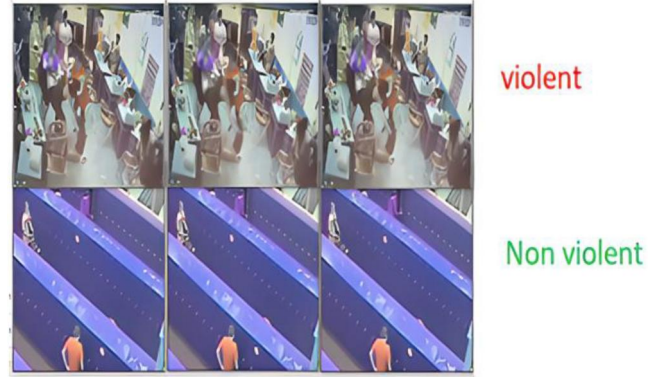
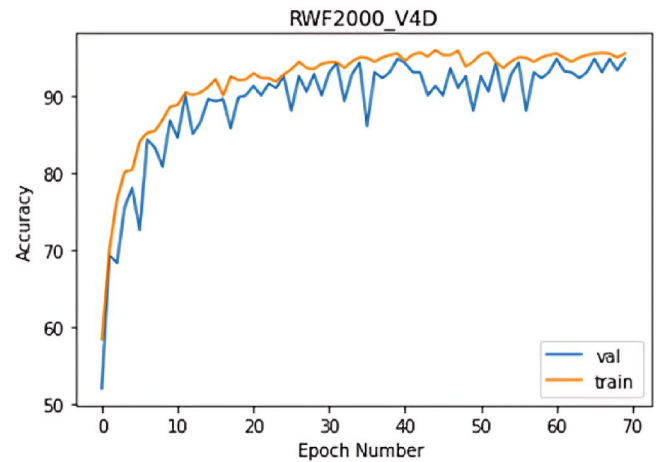
Dataset	Number of clips	Length/clip (s)	Resolution	Scenario	Number of violent clips	Number of non-violent clips
RWF2000 [2]	2000	5	Variable	Surveillance	1000	1000
Hockey Fight [3]	1000	1.6–1.96	360 × 288	Hockey games	500	500
Crowd violence [4]	246	1.04–6.52	Variable	Natural	123	123
Movie fight [45]	200	1.6–2	720 × 480	Movie	100	100

included in violence datasets. Only inside settings, especially an ice hockey rink, are shown in the Hockey Fights dataset. The sequences in the Movies dataset alternate between inside and outside settings. However, none of them portray bad weather. The Crowd dataset looks at large violence that occurs exclusively outside.

Rain, fog, and snow are among the meteorological conditions seen in certain scenes. The RWF2000 dataset depicts a wide range of indoor and outdoor settings, including the street, different athletic event sites, different rooms in a house, and music performance stages, as well as various severe weather conditions.

First RWF2000 has been divided into 70% for training 1400 videos, 700 violent and 700 non-violent, and 15% for validation 300 videos, 150 for violent and 150 for non-violent and 15% for testing. Different experiments have been done with different parameters on the RWF2000 dataset. For model initialisation, ImageNet pre-trained weights were utilised. To begin, each video has been divided into U action units, each of which comprised a number of frames. At first, each frame has been resized to 320×256 and then cropped into 256×256 according to the region of interest resulted from using Farneback's [41] technique as shown in Figure 5 which gives samples of output. After that, Stochastic Gradient Descent (SGD) optimiser has been used with different learning rates, and weight decay is applied with different momentum. The model was trained for 70 epochs. Due to resource constraints, the model was first trained on the 3D ResNet50 backbone with 8-frame inputs, and then the 3D ResNet50 weights were transferred to V4D ResNet50, with all 4D Blocks set to zero. The best result achieved, which is 94.67% for testing after many experiments when each action unit has been set to contain 8 frames and number of action units = 4, so we had 8×4 frames for each video for training with SGD optimiser and learning rate = 0.001 with weight decay = 5×10^{-3} and momentum of 0.9. Violence 4D ResNet50 is then fine-tuned using 8×4 input frames. In the end, optimisation to all 4D Blocks is applied, and train the V4D with 8×4 frames. Figure 6 shows training and validation diagram results. Figure 7 presents the confusion matrix for the RWF2000 test dataset, which included 143 true positives (TP), 9 true negatives (TN), 141 false positives (FP), and 7 false negatives (FN) of the total testing videos.

The proposed method was applied with the parameters which achieved the highest result in RWF2000 on three benchmarks Crowd violence, Movie, and Hockey Fight. Each dataset was split into 70% training, 15% validation, and 15% testing. Using Violence 4D attains accuracy rates of 100%,

**FIGURE 5** Samples of the region of interest of violent and non-violent using flow gates network on RWF2000 dataset.**FIGURE 6** Training and Validation accuracies using RWF2000 with Violence 4D.

97.29%, and 100% in Hockey Fight, Crowd violence, and Movie datasets respectively, as shown in Figure 8. On the Hockey dataset, Figure 9 shows the training accuracy of 100% and the validation accuracy of 100%. Figure 10 presents the confusion matrix for the Hockey test dataset which included 75 FP, 0 FN, 75 TP, and 0 TN. Figure 11 shows the Crowd violence dataset's training and validation accuracies, which are 99.625% and 98% respectively. Figure 12 presents the confusion matrix for Crowd violence test dataset which included 37 FP, 2 FN, 35 TP, and 0 TN. In Figure 13, training and validation accuracies for the Movie dataset are 100% and 100% respectively. Figure 14 presents the confusion matrix for the Movie test dataset which included 30 FP, 0 FN, 30 TP,

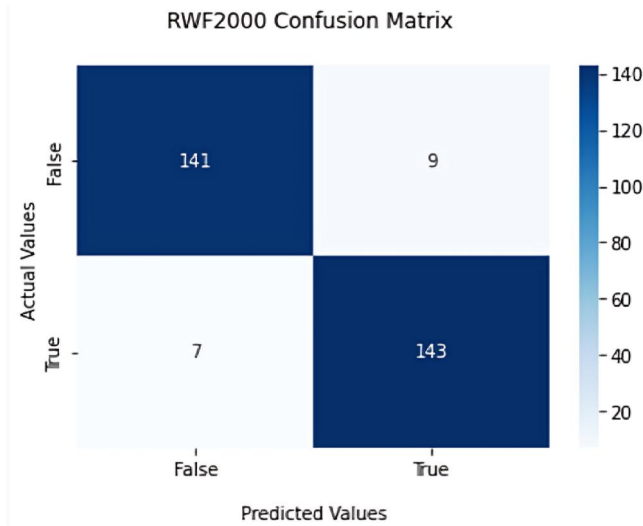


FIGURE 7 Confusion matrix of RWF2000 dataset on test data.

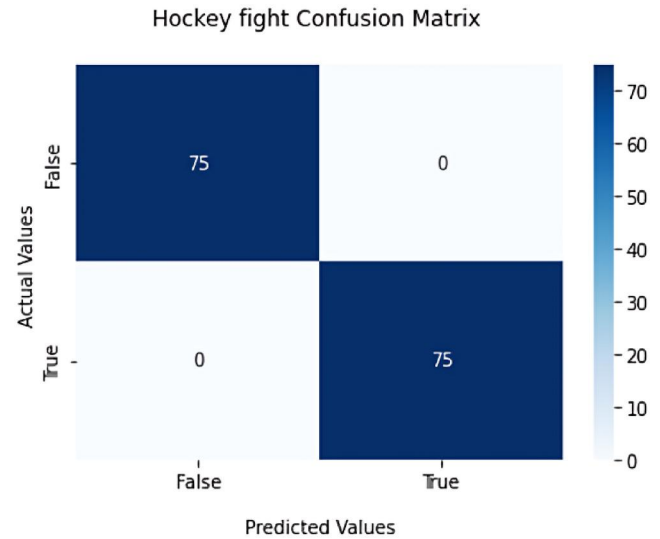


FIGURE 10 Confusion matrix of Hockey fight dataset on test data.

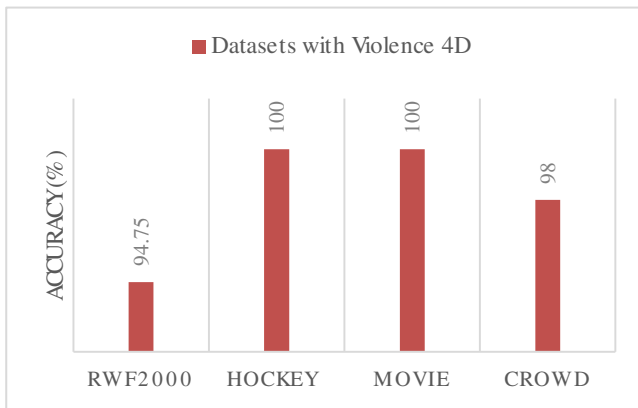


FIGURE 8 Accuracies on the four violence benchmark datasets using violence 4D.

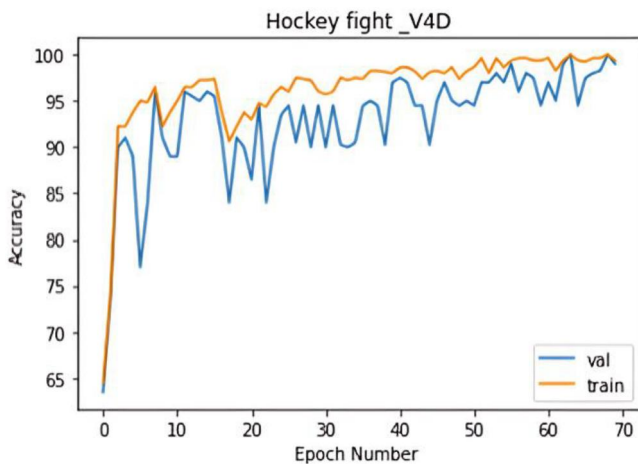


FIGURE 9 Training and validation accuracies using Hockey dataset with violence 4D.

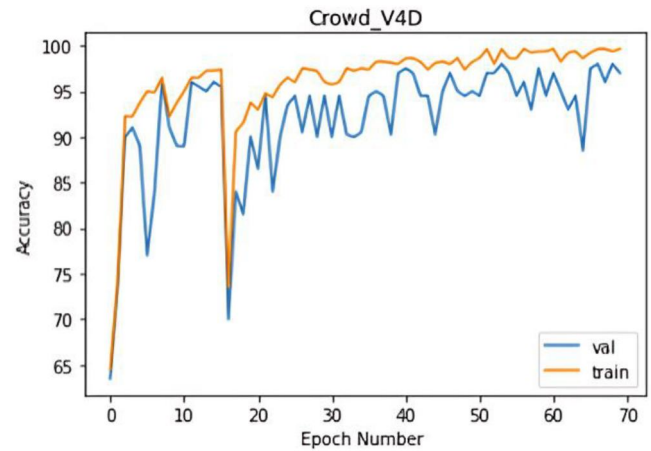


FIGURE 11 Training and validation accuracies using Crowd violence with Violence 4D.

and 0 TN. The results of four violent datasets resulting from each confusion matrix are shown in Table 2 in terms of accuracy, precision, recall, f1 score, and specificity. In conclusion, the proposed approach showed promising results when evaluated against a challenging dataset gathered from real-world data.

4.3 | Comparison between violence detection methods and violence 4D on different datasets

The effectiveness of various violence detection algorithms and our violence 4D model, which were tested on the RWF2000, Crowd violence, Movie, and Hockey fight datasets, are compared in Table 3. When comparing our violence 4D method to those already in existence, we can see that Movie

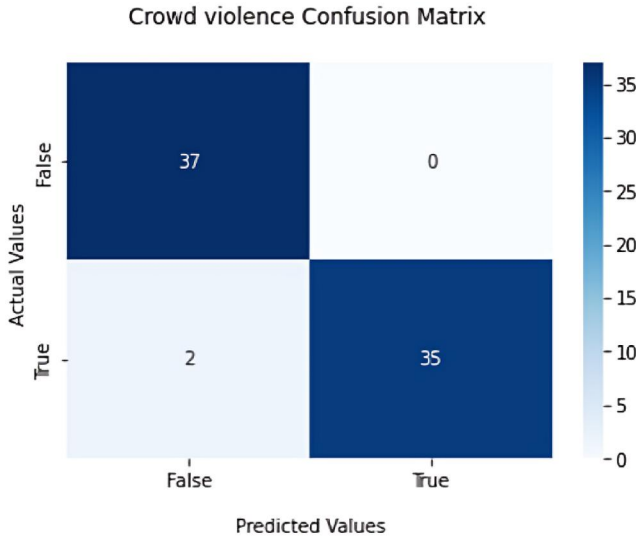


FIGURE 12 Confusion matrix of Crowd violence dataset on test data.

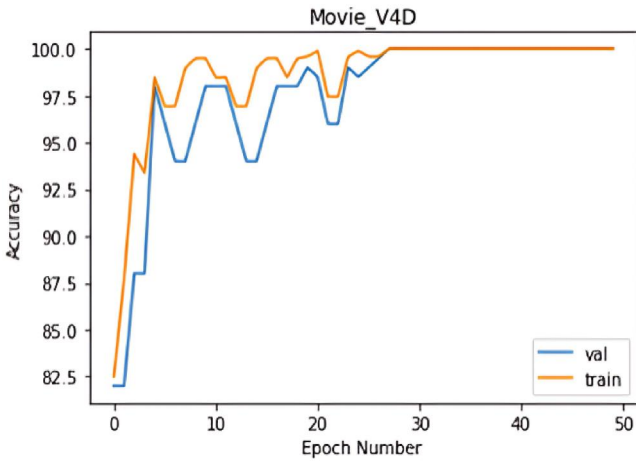


FIGURE 13 Training and validation accuracies using Movie dataset with Violence 4D.

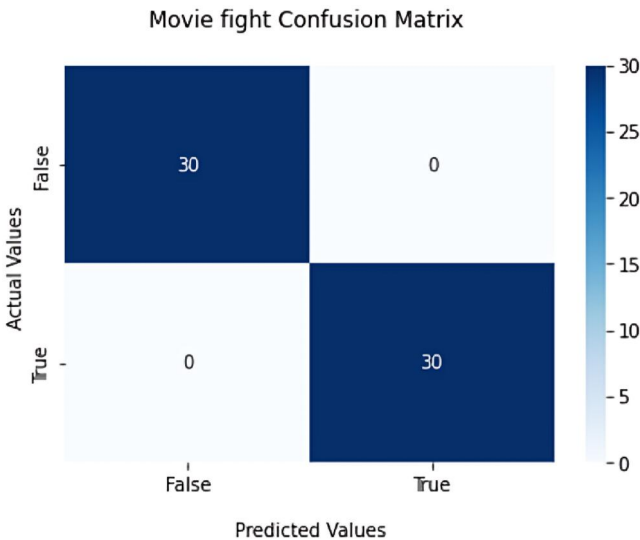


FIGURE 14 Confusion matrix of Movie dataset on test data.

TABLE 2 Datasets results using confusion matrix

Dataset	Accuracy	Precision	Recall	F1_Score	Specificity
RWF2000	0.9467	0.9407	0.9533	0.9400	0.9470
Hockey fight	1.0000	1.0000	1.0000	1.0000	1.0000
Crowd violence	0.9729	1.0000	0.9459	1.0000	0.9722
Movie fight	1.0000	1.0000	1.0000	1.0000	1.0000

dataset achieved similar accuracy results and Crowd violence achieved a higher result than violence 4D, as we obtained the second-best result; but models used are much heavier than ours, like Flow Gated Net [45] as using two-stream convolution neural networks is challenging and expensive due to parallel optimisation. 2D CNNs [48] are employed for learning from clip-level features rather than learning from the entire video; they are used for feature extraction from frame level. In terms of accuracy, our approach can generally compete with other methods. The RWF2000 produced the best results.

Our method is accurate and computationally efficient because we combined four-dimensional residual blocks with three-dimensional convolutional neural networks to learn both long-term and short-term spatiotemporal representations.

4.4 | Ablation study

In this section, the influence of different components and parameters in Violence 4D is being discussed. All experiments were performed on the RWF2000 dataset. To initialise the model, pre-trained weights have been employed from ImageNet. After many experiments using Violence 4D (ResNet50) as shown in Table 4; The SGD optimiser is used with a learning rate of 0.001, weight decay is set to 5×10^{-4} , and a momentum of 0.9. The model is trained for 70 epochs in total.

4.4.1 | Sampling method

The video sampling method will affect the network performance. A clip of 32 frames is randomly chosen from each part after evenly dividing the entire clip into U parts. Then uniformly sample four frames with a fixed stride of eight to generate an action unit for each clip. Violence 4D performance is compared using two different sampling strategies: sparse sampling in Ref. [39] and dense sampling. With two different models as the backbone: ResNet-18 and ResNet-50.

Dense sampling: a fixed interval θ is used to select frames from each U part. The length of sampling is set to 32; then, the value of θ is set from 1 to $T/32$. When $\theta = 1$, 32 consecutive frames from each U part are selected.

Sparse sampling: When $\theta = T/32$, sampling includes using the maximum interval to select the 32 video frames. The recognition results of these two methods are shown in Table 5.

From these results, by using different backbone networks, when employing sparse sampling with (ResNet-18) and

TABLE 3 Comparison between methods and Violence 4D on different datasets

Method	RWF2000 dataset (%)	Crowd violence dataset (%)	Movie dataset (%)	Hockey Fight dataset (%)
ViF [5]	—	81.30	—	82.90
ConvLSTM [24]	—	94.57	100.00	97.10
FightNet [32]	—	—	100.00	97.00
3D ConvNet [42]	—	94.30	99.97	99.62
C3D [43]	—	84.44	100.00	96.50
I3D (RGB only) [44]	—	86.67	100.00	98.50
I3D (flow only) [44]	—	88.89	100.00	84.00
I3D (fusion) [44]	—	88.89	100.00	97.50
Flow gated net [45]	87.25	88.87	100.00	98.00
SepConvLSTM-M [46]	89.75	—	100.00	99.00
SepConvLSTM-A [46]	87.75	—	100.00	99.00
SepConvLSTM-C [46]	89.25	—	100.00	99.50
SPIL convolution [47]	89.30	94.50	98.50	—
2D CNNs + LSTM [48]	92.00	98.00	100.00	99.60
2D spatio-temporal representations [49]	93.80	90.60	99.50	94.40
Violence 4D	94.67	97.29	100.0	100.0

Note: First bold value is the highest achieved accuracy in RWF200 dataset using violence 4D. Second bold value is the highest achieved accuracy in Crowd violence dataset using 2DCNNs + LSTM. Third and fourth bold values are the highest achieved accuracy in Movie and Hockey Fight datasets using violence 4D.

Abbreviations: CNN, convolution neural network; LSTM, long short-term memory.

TABLE 4 Comparison results of the different parameters

Momentum	Optimiser	Learning rate	Weight decay	Accuracy (%)
0.7	SGD	0.010	5×10^{-4}	84.25
		0.001	5×10^{-3}	85.50
0.8	SGD	0.010	5×10^{-4}	86.42
		0.001	5×10^{-3}	88.50
0.9	SGD	0.010	5×10^{-4}	92.25
		0.001	5×10^{-3}	94.75

Abbreviation: SGD, stochastic gradient descent.

TABLE 5 Comparison results of the different sampling methods

Method	Dense sampling (%)	Sparse sampling (%)
Violence 4D (ResNet-18)	87.53	92.40
Violence 4D (ResNet-50)	89.63	94.75

(ResNet-50), the accuracy of violence recognition is improved by 4.87% and 5.12% respectively.

4.4.2 | Cropping method

Each frame is resized to 320×256 and then cropped. Performance of cropping which has been compared between two methods random cropping [50] and optical flow [40]. Table 6

TABLE 6 Comparison results of the different cropping methods

Method	Violence 4D (ResNet-18) (%)	Violence 4D (ResNet-50) (%)	Time (s) for each clip
Random cropping	87.53	92.4	1.0667
Optical flows	89.63	94.75	1.7780

TABLE 7 Comparison results of the different backbone networks

Backbone	Without violence 4D (%)	With violence 4D (%)
ResNet18	85.7	92.40
ResNet50	87.63	94.75

shows that, by using two different backbone networks, when using optical flow [41] with Violence 4D (ResNet-18) and Violence 4D (ResNet-50) accuracy improved by 3.1% and 3.93% respectively. Although optical flows achieved higher accuracy but consumed more resources and time in processing as it consumed 0.7113 ms per video more than random cropping.

4.4.3 | Backbone networks

The accuracy of the Violence 4D changes with different backbone networks, as shown in Table 5. To demonstrate that the Violence 4D is robust for the different backbone

networks, Table 7 compares the findings of different backbone networks with the Violence 4D. With the Violence 4D, the accuracy of backbone networks improved by 6.7% on ResNet18 and 7.12% on ResNet50. According to these findings, Violence 4D can increase accuracy for various backbone networks.

4.4.4 | Four-dimensional convolution kernels with different backbone networks

Convolution in four dimensions Kernels employ three types of forms: $k \times k \times k \times k$, $k \times k \times 1 \times 1$ and $k \times 1 \times 1 \times 1$. When k is = 3 then a single 4D block is applied at the end of res4 in ResNet18 and ResNet50. As shown in Table 8, Violence 4D with $3 \times 3 \times 3 \times 3$ kernel can achieve the highest performance. The kernel of $3 \times 3 \times 1 \times 1$ is used because of the cost benefit analysis between performance and parameters of model.

4.4.5 | Four-dimensional convolution blocks

Using four-dimensional Blocks for Violence 4D, the effect of number and position has been examined. Violence 4D performance is evaluated by employing a single $3 \times 3 \times 1 \times 1$ 4D block at res3, res4, and res4 with res5. Table 9 demonstrates that applying the four-dimensional blocks at res3, res4 results in good accuracy, putting one 4D block at res3 and one at res4 improved accuracy because of the combination of short- and long-term properties which provide more meaningful representation. The input size shape when one 4D block 1 at res 3, 1 at res 4 is $4 \times 4 \times 256 \times 256$.

TABLE 8 Comparison results of the different backbone networks with different shapes of 4D convolution kernel

Model	Shape of kernel	Accuracy (%)
Violence 4D (ResNet18)	$3 \times 1 \times 1 \times 1$	91.75
Violence 4D (ResNet18)	$3 \times 3 \times 1 \times 1$	92.10
Violence 4D (ResNet18)	$3 \times 3 \times 3 \times 3$	92.60
Violence 4D (ResNet50)	$3 \times 1 \times 1 \times 1$	93.10
Violence 4D (ResNet50)	$3 \times 3 \times 1 \times 1$	93.65
Violence 4D (ResNet50)	$3 \times 3 \times 3 \times 3$	94.21

TABLE 9 Comparison results of the different position and number of 4D blocks

Model	4D kernel	Accuracy (%)
Violence 4D (ResNet50)	1 at res3	93.10
Violence 4D (ResNet50)	1 at res4	93.65
Violence 4D (ResNet50)	1 at res3, 1 at res4	94.75

4.4.6 | Number of action units U

Violence 4D has been training with different numbers of action units U. Residual 4D block with $3 \times 3 \times 1 \times 1$ kernel is added at the end of res3 and res4 of ResNet50. Table 10 shows that U does not have a major influence on performance, implying that: Violence 4D is a learning model for Violence features of video-level which is resistant to the number of short-term units. In general action does not have many steps, as a result, raising U is ineffective. Furthermore, increasing the number of action units requires that the fourth dimension to be increased, which needs bigger 4D kernel to cover the long-range evolution of spatial-temporal representation.

ResNet50 is employed as the backbone for Violence 4D, according to ablation study. Units of action are set to 4, with 8 frames from each unit, for a total of 4×8 frames for training. We train the model in phases due to computational resource constraints. We begin by training the ResNet50 backbone using 8-frame then ResNet50 weights are then loaded into Violence 4D, with all 4D Blocks set to zero. After that, the Violence 4D is fine-tuned using 8×4 input frames. Finally, all 4D Blocks are optimised.

5 | DISCUSSION

As shown in Tables 2 and 3, the proposed model achieved higher accuracy than state-of-the-art compared methods. We attribute this superior performance to its use of four-dimensional residual blocks with 3D CNNs (ResNet50). This combination helps learn long-term and short-term spatio-temporal representations as opposed to using only 3D CNNs which look only at a portion of the overall video. 3D CNNs frequently have a high number of model parameters, necessitating a significant quantity of training data to obtain acceptable results. Also, using two-stream convolution neural networks is difficult and expensive because of parallel optimisation. The proposed model has been tested on four violence benchmarks that are often used in video violence detection: RWF2000, Crowd violence, Hockey Fight, and Movie dataset. The best results have been achieved on RWF2000, which was a challenging dataset as it is the largest surveillance violence video dataset till now. We also

TABLE 10 Comparison results of the different number of action units

Model	Number of action units	Number of frames	Accuracy
Violence 4D (ResNet50)	2	2	94.50
Violence 4D (ResNet50)	4	4	94.62
Violence 4D (ResNet50)	4	8	94.75

achieved the highest accuracy on the Hockey Fight dataset. While there are other video datasets for violence identification; they suffer from limitations such as small scale, limited variety, and low imaging resolution and they were not used in this paper.

6 | CONCLUSION AND FUTURE WORK

In this paper, a short-term and long-term spatiotemporal representations using video 4D convolution neural network (V4D) is proposed. The model employs resnet50 as its backbone and dense optical flow to get the region of interest of RGB frames. Because video four-dimensional convolution neural networks block records inter-clip interactions, they can improve the representation capability of three-dimensional convolution neural networks at the clip level; 4D residual blocks are hierarchically merged with current 3D CNNs for long-term modelling. As a possible future enhancement to the proposed model, optical flow channels are applied with RGB channels, and they are merged using two streams with video four-dimensional convolution.

AUTHOR CONTRIBUTIONS

Mai Magdy: Conceptualisation; Data curation; Formal analysis; Investigation; Methodology; Project administration; Resources; Software; Validation; Visualisation; Writing – original draft; Writing – review & editing. **Mohamed Waleed Fakhr:** Investigation; Project administration; Supervision; Validation; Visualisation. **Fahima A. Maghraby:** Investigation; Project administration; Supervision; Validation; Visualisation.

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CONFLICT OF INTEREST

The authors declared that they have no conflicts of interest to this work.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in [Movies Fight Detection Dataset] at [<https://academictorrents.com/details/70e0794e2292fc051a13f05ea6f5b6c16f3d3635>] reference number [3]. The data that support the findings of this study are openly available in [Hockey Fight Detection Dataset] at [<https://paperswithcode.com/dataset/hockey-fight-detection-dataset>] reference number [4]. The data that support the findings of this study are openly available in [Crowd Violence] [<https://www.openu.ac.il/home/hassner/data/violentflows/>] reference number [5]. The data that support the findings of this study are available from [Ming Cheng]. Restrictions apply to the availability of these data, which were used under license for this study. Data are [<https://github.com/mchengny/RWF2000-Video-Database-for-Violence-Detection>] with the permission of [Ming Cheng], reference number [45].

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